

2024, Mar. 2024 Release LOC:Acute Adult

Infection: Cellulitis

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Select Day

Initial review ^(1, 2, 3, 4)Episode Day 1 ^(1, 3, 4)Episode Day 2 ^(3, 4, 31)Episode Day 3 ^(4, 35, 36)Episode Day 4 ^(4, 36)

Episode Day 5

Discharge Screens ⁽⁴⁰⁾

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is for patients with a cellulitis infection. For patients with skin infections including: abscess, cutaneous, herpes zoster, wound infection, surgical site infection, and diabetic foot ulcers, refer to Infection: Skin subset. For patients who begin to exhibit signs and symptoms consistent with sepsis and need additional interventions which are not provided in the current subset, please refer to the Infection: Sepsis subset.

Introduction:

Cellulitis is a diffuse spreading skin infection that extends to the dermis and subcutaneous fat. Clinical manifestations can range from mild to severe and include areas of edema, erythema (redness), warmth, and are often accompanied by lymphangitis. Systemic manifestations can be life-threatening and when symptoms of fever, tachycardia, confusion, hypotension, or leukocytosis are present, it may be indicative of a severe infection. The most frequent organisms causing cellulitis are β -hemolytic streptococci and *Staphylococcus aureus* (Boettler et al., Am J Clin Dermatol 2022, 23: 153-65; Rrapi et al., Med Clin North Am 2021, 105: 723-35).

Evaluation and treatment:

Most patients with cellulitis can be managed in the outpatient setting with oral anti-infectives and wound management. Hospitalization may be required if the cellulitis fails to respond to outpatient anti-infective treatment, the patient is currently using IV substances, has severe obesity (Body mass index (BMI) greater than 40kg/m²) and unable to manage wound care (e.g., unable to offload, unable to reach or dress wound), or has comorbid conditions (e.g., lymphedema, immunocompromised) associated with poor outcomes.

Intravenous anti-infective treatment for cellulitis is based on history of present illness, risk for resistant organisms, and physical assessment. Specialty consultations with dermatology and infectious disease are recommended if the cellulitis infection is not responding or slow to respond to anti-infective treatment (Boettler et al., Am J Clin Dermatol 2022, 23: 153-65; Rrapi et al., Med Clin North Am 2021, 105: 723-35; Ko et al., JAMA Dermatol 2018, 154: 529-36; Stevens et al., Clin Infect Dis 2014, 59: e10-52).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites,

and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Boettler et al., Am J Clin Dermatol 2022, 23: 153-65; Rrapi et al., Med Clin North Am 2021, 105: 723-35; Stevens et al., Clin Infect Dis 2014, 59: e10-52).

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3, 4)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, Both:** ⁽⁵⁾● Finding, ≥ **One:**

- Animal or human bite of the face, hand, or genitalia
- BMI ≥ 40 kg/m² and unable to manage wound care ⁽⁶⁾

● High risk for complication, **All:**

- Active IV substance use disorder and discharge planning ⁽⁷⁾
- Patient unreliable and unable to adhere to outpatient instructions or treatment ⁽⁸⁾
- Caregiver unavailable or unable to manage care

● Failed outpatient anti-infective treatment (includes PO), ≥ **One:** ⁽⁹⁾

- Continued deterioration despite ≥ 24h treatment (includes PO) ⁽¹⁰⁾
- Non-compliant with outpatient anti-infective (includes PO)

● Unresponsive to outpatient anti-infective (includes PO) and, ≥ **One:** ⁽¹¹⁾

- ≥ 3d
- ≥ 5 doses

- Lymphedema ⁽¹²⁾

- Multifocal cellulitis ⁽¹³⁾

- Anti-infective ⁽¹⁴⁾

● **ACUTE, Both:** ⁽¹⁵⁾● Finding, ≥ **One:**

- Immunocompromised ⁽¹⁶⁾
- Located over a prosthesis or implanted device ⁽¹⁷⁾
- Orbital cellulitis confirmed by imaging ^(18, 19, 20, 21)

- Anti-infective ⁽¹⁴⁾

● **INTERMEDIATE, ≥ One:** ⁽²²⁾

- NIPPV ⁽²³⁾

- Oxygen ≥ 40%(0.40), ≤ 2d ⁽²⁴⁾

● **CRITICAL, ≥ One:** ⁽²⁵⁾

- DIC ⁽²⁶⁾

- ECMO or ECLS ⁽²⁷⁾

● Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** ⁽²⁸⁾

● Hypotension, ≥ **One:**

- MAP < 65 mmHg
- Systolic blood pressure < 90 mmHg and < baseline

● Finding, ≥ **One:**

- Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽²⁹⁾
- Urine output < 0.5 mL/kg/h

- Hemodynamic monitoring, invasive ⁽³⁰⁾

- Mechanical ventilation

● NIPPV and, ≥ **One:** ⁽²³⁾

- Arterial Po₂ ≤ 39 mmHg(5.2 kPa)

● Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**

- COPD and arterial or venous pH ≤ 7.24
- No hx of COPD

- Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 1, One:** ^(1, 3, 4)

(Symptom or finding within 24h)

● **OBSERVATION, Both:** ⁽⁵⁾

- Finding, **≥ One:**
 - Animal or human bite of the face, hand, or genitalia
 - BMI $\geq 40 \text{ kg/m}^2$ and unable to manage wound care ⁽⁶⁾
- High risk for complication, **All:**
 - Active IV substance use disorder and discharge planning ⁽⁷⁾
 - Patient unreliable and unable to adhere to outpatient instructions or treatment ⁽⁸⁾
 - Caregiver unavailable or unable to manage care
- Failed outpatient anti-infective treatment (includes PO), **≥ One:** ⁽⁹⁾
 - Continued deterioration despite $\geq 24\text{h}$ treatment (includes PO) ⁽¹⁰⁾
 - Non-compliant with outpatient anti-infective (includes PO)
- Unresponsive to outpatient anti-infective (includes PO) and, **≥ One:** ⁽¹¹⁾
 - $\geq 3\text{d}$
 - ≥ 5 doses
- Lymphedema ⁽¹²⁾
- Multifocal cellulitis ⁽¹³⁾
- Anti-infective ⁽¹⁴⁾
- **ACUTE, Both:** ⁽¹⁵⁾
 - Finding, **≥ One:**
 - Immunocompromised ⁽¹⁶⁾
 - Located over a prosthesis or implanted device ⁽¹⁷⁾
 - Orbital cellulitis confirmed by imaging ^(18, 19, 20, 21)
 - Anti-infective ⁽¹⁴⁾
- **INTERMEDIATE, $\geq \text{One}$:** ⁽²²⁾
 - NIPPV ⁽²³⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2\text{d}$ ⁽²⁴⁾
- **CRITICAL, $\geq \text{One}$:** ⁽²⁵⁾
 - Necrotizing fasciitis (**use General Surgical subset**)
 - DIC ⁽²⁶⁾
 - ECMO or ECLS ⁽²⁷⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24\text{h}$ and, **Both:** ⁽²⁸⁾
 - Hypotension, **≥ One:**
 - MAP $< 65 \text{ mmHg}$
 - Systolic blood pressure $< 90 \text{ mmHg}$ and $< \text{baseline}$
 - Finding, **≥ One:**
 - Lactic acid $> 2 \text{ mmol/L}(18.0 \text{ mg/dL})$ ⁽²⁹⁾
 - Urine output $< 0.5 \text{ mL/kg/h}$
 - Hemodynamic monitoring, invasive ⁽³⁰⁾
 - Mechanical ventilation
 - NIPPV and, **≥ One:** ⁽²³⁾
 - Arterial $\text{Po}_2 \leq 39 \text{ mmHg}(5.2 \text{ kPa})$
 - Arterial or venous $\text{Pco}_2 \geq 60 \text{ mmHg}(8.0 \text{ kPa})$ and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
 - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2, One:** ^(3, 4, 31)
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **Both:**
 - Anti-infective regimen established and tolerated
 - Cellulitis resolving
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
 - Cellulitis unresolved and anti-infective ⁽³²⁾
 - **ACUTE, $\geq \text{One}$:**

- Cellulitis unresolved and anti-infective ⁽³²⁾
- Adverse anti-infective drug reaction requiring change in medication $\leq 24\text{h}$ ⁽³³⁾
- Post Critical care $\leq 24\text{h}$
- **INTERMEDIATE, $\geq$ One:**
 - NIPPV ⁽²³⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2\text{d}$ ⁽²⁴⁾
- **CRITICAL, $\geq$ One:**
 - DIC ⁽²⁶⁾
 - ECMO or ECLS ⁽²⁷⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24\text{h}$ and, **Both:** ⁽²⁸⁾
 - Hypotension, $\geq$ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, $\geq$ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽²⁹⁾
 - Urine output < 0.5 mL/kg/h
 - Hemodynamic monitoring, invasive ⁽³⁰⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽²³⁾
 - $\text{FiO}_2 > 50\%(0.50)$
 - Respiratory intervention every 1-2h ⁽³⁴⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3, One:** ^(4, 35, 36)
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **Both:**
 - Anti-infective regimen established and tolerated
 - Cellulitis resolving
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
 - For cellulitis use Episode Day 3 at a higher level of care
 - For a condition other than cellulitis use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁷⁾
 - Anti-infective regimen established and tolerated
 - Cellulitis resolving
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽³⁸⁾
 - **Complication or comorbidity and clinically stable for discharge, One:**
 - Adverse drug reaction resolving
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Cellulitis unresolved, **Both:** ⁽³²⁾
 - Inflammation worsening or unimproved ⁽³⁹⁾
 - Anti-infective ⁽³²⁾
 - Adverse anti-infective drug reaction requiring change in medication $\leq 24\text{h}$ ⁽³³⁾
 - Post Critical care $\leq 24\text{h}$
 - **INTERMEDIATE, $\geq$ One:**
 - NIPPV ⁽²³⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2\text{d}$ ⁽²⁴⁾
 - **CRITICAL, $\geq$ One:**
 - DIC ⁽²⁶⁾
 - ECMO or ECLS ⁽²⁷⁾

- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, **Both:** ⁽²⁸⁾
 - Hypotension, $\geq$ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, $\geq$ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽²⁹⁾
 - Urine output < 0.5 mL/kg/h
 - Hemodynamic monitoring, invasive ⁽³⁰⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽²³⁾
 - FiO₂ $> 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽³⁴⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 4, One:** ^(4, 36)
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁷⁾
 - Anti-infective regimen established and tolerated
 - Cellulitis resolving
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽³⁸⁾
 - **Complication or comorbidity and clinically stable for discharge, One:**
 - Adverse drug reaction resolving
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - *C. difficile* infection, hospital acquired (**use Infection: GI/GYN subset**)
 - **Cellulitis unresolved, Both:** ⁽³²⁾
 - Inflammation worsening or unimproved ⁽³⁹⁾
 - Anti-infective ⁽³²⁾
 - Adverse anti-infective drug reaction requiring change in medication ≤ 24 h ⁽³³⁾
 - Post Critical care ≤ 24 h
 - **INTERMEDIATE, $\geq$ One:**
 - NIPPV ⁽²³⁾
 - Oxygen $\geq 40\%$ (0.40), ≤ 2 d ⁽²⁴⁾
 - **CRITICAL, $\geq$ One:**
 - DIC ⁽²⁶⁾
 - ECMO or ECLS ⁽²⁷⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, **Both:** ⁽²⁸⁾
 - Hypotension, $\geq$ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, $\geq$ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽²⁹⁾
 - Urine output < 0.5 mL/kg/h
 - Hemodynamic monitoring, invasive ⁽³⁰⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽²³⁾
 - FiO₂ $> 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽³⁴⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
 - **Episode Day 5, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁷⁾

- Anti-infective regimen established and tolerated
- Cellulitis resolving
- **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽³⁸⁾
 - **Complication or comorbidity and clinically stable for discharge, One:**
 - Adverse drug reaction resolving
- **Non-responder, not clinically stable for discharge and does not meet partial responder criteria, One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
- **INTERMEDIATE, One:**
 - **Non-responder, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
- **CRITICAL, One:**
 - **Non-responder, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, One:**
 - **Use Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, One: ⁽⁴⁰⁾

- **HOME, Both:**
 - **Level of care appropriateness, Both:**
 - Home environment safe and accessible ⁽⁴¹⁾
 - Patient or caregiver demonstrates ability to manage care
 - **Cellulitis discharge planning, All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁴²⁾
 - **Education on the importance of, All:**
 - Notifying healthcare provider of fever, chills, increased swelling, redness, or discharge of the infected area
 - Strict medication adherence
 - Keeping the infected area clean
 - Medication reconciliation ⁽⁴³⁾
 - Follow-up care planned ⁽⁴⁴⁾
 - Identify and address transportation needs
- **HOME CARE, Both:**
 - **Level of care appropriateness, All:**
 - Home environment safe and accessible ⁽⁴¹⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(45, 46)
 - **Skilled services, ≥ One:** ⁽⁴⁷⁾
 - Skilled nursing ⁽⁴⁸⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁴⁹⁾
 - **Cellulitis discharge planning, All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁴²⁾
 - **Education on the importance of, All:**
 - Notifying healthcare provider of fever, chills, increased swelling, redness, or discharge of the infected area
 - Strict medication adherence
 - Keeping the infected area clean
 - Medication reconciliation ⁽⁴³⁾
 - Follow-up care planned ⁽⁴⁴⁾

- Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽⁵⁰⁾
 - Skilled treatment, **≥ One:**
 - Clinical assessment ⁽⁵¹⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁵²⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 1\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $1\text{-}2\text{x}/24\text{h}$ ⁽⁴⁸⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(53, 54)
 - Functional impairments requiring at least supervision ⁽⁵⁵⁾
 - Able to tolerate 1h/d of skilled therapy $\geq 5\text{d/wk}$
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁴³⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 2\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $4\text{h}/24\text{h}$ ⁽⁴⁸⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(53, 54)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵⁵⁾
 - Able to tolerate $2\text{-}3\text{h/d}$ of skilled therapy $\geq 5\text{d/wk}$ and ≥ 1 therapy discipline
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁴³⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(53, 54)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵⁵⁾
 - Able to tolerate $2\text{-}3\text{h/d}$ of skilled therapy $\geq 5\text{d/wk}$ and ≥ 2 therapy disciplines
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁴³⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(53, 54)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵⁵⁾
 - Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁵⁶⁾
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁴³⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level
 - **In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁵⁷⁾
 - Ventilator dependent $\geq 6h/d$ and weaning planned ^(58, 59)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁶⁰⁾
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁴³⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
121 ACUTE MAJOR EYE INFECTIONS WITH CC/MCC	n/a	4.4	CMS GMLOS
122 ACUTE MAJOR EYE INFECTIONS WITHOUT CC/MCC	n/a	3	CMS GMLOS
154 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITH MCC	n/a	4.2	CMS GMLOS
155 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITH CC	n/a	2.9	CMS GMLOS
156 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
573 SKIN GRAFT FOR SKIN ULCER OR CELLULITIS WITH MCC	n/a	12.2	CMS GMLOS
574 SKIN GRAFT FOR SKIN ULCER OR CELLULITIS WITH CC	n/a	8	CMS GMLOS
575 SKIN GRAFT FOR SKIN ULCER OR CELLULITIS WITHOUT CC/MCC	n/a	4.4	CMS GMLOS
602 CELLULITIS WITH MCC	n/a	4.8	CMS GMLOS
603 CELLULITIS WITHOUT MCC	n/a	3.2	CMS GMLOS
727 INFLAMMATION OF THE MALE REPRODUCTIVE SYSTEM WITH MCC	n/a	4.9	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
728 INFLAMMATION OF THE MALE REPRODUCTIVE SYSTEM WITHOUT MCC	n/a	2.9	CMS GMLOS
757 INFECTIONS, FEMALE REPRODUCTIVE SYSTEM WITH MCC	n/a	5	CMS GMLOS
758 INFECTIONS, FEMALE REPRODUCTIVE SYSTEM WITH CC	n/a	3.7	CMS GMLOS
759 INFECTIONS, FEMALE REPRODUCTIVE SYSTEM WITHOUT CC/MCC	n/a	2.6	CMS GMLOS
CELLULITIS	27%	3.9	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Cellulitis, expected progress within 24-48h:

- Vital signs improving
- Laboratory values normalizing
- Erythema and edema improving or not increasing
- Activity increasing
- Pain minimal or pain free

Consider:

- Changing antibiotic route from intravenous (IV) to PO
- Discharge planning

4:

Cellulitis, care facilitation:

Potential risk of readmission

- Knowledge deficit or non-adherence with condition management identified (**Home Care, Readmission reduction team**)
- Co-morbid chronic obstructive pulmonary disease or heart failure (**Home Care, Readmission reduction team**)

Functional or activity of daily living (ADL) impairment - consider an alternate level of care for therapy needs

- Physical therapy provided in the home setting should be considered if the home is safe and accessible (**Home Care**)
- If home environment is unsafe, consider a skilled nursing facility (**SNF**)

Wound care

- Wound care in the home setting should be considered if the patient requires daily assessment, intervention, or evaluation of a wound (**Home Care**)
- If home environment is not appropriate due to clinical complexity or patient safety, considered a skilled nursing facility (**SNF**)

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end-stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent)

supplements)

- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meal programs, homemaking services, and transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., *Pediatr Rev* 2018, 39: 235-46)

Functional or ADL impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment non-adherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). *Psychosis and schizophrenia in adults: prevention and management*. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication non-adherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure, information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies (e.g., blood lead level testing and follow up) (Centers for Disease Control and Prevention, *Lead Poisoning Prevention*. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance information should be given after application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., *CMAJ* 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (e.g., serious mental illness, history of hospitalizations, functional impairment), consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., *PLoS One* 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

5:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care

services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

6:

The literature recognizes excessive weight or obesity, defined as a body mass index (BMI) of at least 30 kg/m², as a risk factor for acute or recurrent cellulitis, among other skin infections. Obesity negatively affects skin integrity, micro- and macro-vascularity, and lymphatic drainage and function, predisposing this patient population to an increased risk of skin infections and compromised wound healing. Diabetes also plays a role in heightening the risk of infections as a significant comorbidity. Additionally, in this patient population, antibiotic underdosing based on pharmacokinetics and volume of distribution increases the odds of treatment failure by more than 1.5-fold for every 10 kg/m² in BMI. Due to the hazards of immobility for the patient with obesity (BMI of 40 kg/m² or more; class III obesity), adhering to a recommended wound care regimen may be challenging and lead to poor wound healing. Immobility may be caused by a lack of appropriate bariatric equipment or by limitations due to body habitus (Ewens et al., BMC Nurs 2022, 21: 33; Ong et al., Int J Gen Med 2022, 15: 6561-72; Pai, Clin Pharmacol Ther 2021, 109: 942-51; Ghilotti et al., Int J Epidemiol 2019, 48: 1783-94; Huttunen and Syrjanen, Int J Obes (Lond) 2013, 37: 333-40). **Note:** The BMI used for Observation level of care reflects the opinion of InterQual's® external peer reviewers. The criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied practice and geographic settings.

7:

Patients at high risk for complications due to medication non-adherence or inadequate treatment plans such as those with active intravenous (IV) substance use disorder, may require hospitalization for monitored anti-infective treatment and consultation with infectious disease specialist. While most patients may be successfully treated as outpatient, the decision to hospitalize should be made on an individual basis (Serota et al., Infect Dis Clin North Am 2020, 34: 539-58).

8:

Patient unreliable refers to the following:

- Inability or unwillingness to follow an outpatient monitoring or treatment plan
- Inability to understand the signs of deterioration and recognize the need for prompt medical attention

9:

Instruction: Failed outpatient management refers to patients that have not responded to anti-infective treatment received through the medical practitioner's office, outpatient clinic, or a trial of home care services, and includes care delivered in the Observation setting.

10:

Patients experiencing continued deterioration of their cellulitis may have signs of expansion or spread of cellulitis despite anti-infective treatment (Rrapi et al., Med Clin North Am 2021, 105: 723-35).

11:

A patient should demonstrate objective signs of clinical improvement within 72 hours of initiation of treatment.

12:

Cellulitis that occurs in a patient with chronic lymphedema requires complex treatment because the lymphatic system drainage is impaired and due to poorer peripheral circulation, cannot adequately fight infections. If warranted, obtaining wound cultures may assist in narrowing anti-infective treatment based on culture results (Boettler et al., Am J Clin Dermatol 2022, 23: 153-65; Pulia, Rosen's emergency medicine: concepts and clinical practice, Tenth ed. 2023).

13:

Multifocal cellulitis is a rare condition that may present initially as a single area of cellulitis. Within a few weeks after initial anti-infective treatment, multiple areas of a painful rash/cellulitis develop on the body and can be associated with fever or leukocytosis (Rauseo et al., Clin Infect Dis 2020, 70: 531-4; Pulia, Rosen's emergency medicine: concepts and clinical practice, Tenth ed. 2023).

14:

Intravenous anti-infective treatment for cellulitis is based on history of present illness, risk for resistant organisms, and physical assessment. Specialty consultations with dermatology and infectious disease are recommended if the cellulitis infection is unresponsive or slow to respond to anti-infective treatment (Boettler et al., Am J Clin Dermatol 2022, 23: 153-65; Ko et al., JAMA Dermatol 2018, 154: 529-36).

15:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

16:

Immunocompromised refers to the patient who has a decreased capacity to defend against invading organisms as a result of an impaired immune system due to several comorbidities. The principal manifestation of immunodeficiency is increased susceptibility to infection with unusual severity, frequency, and duration which differs from that of an individual who is immunocompetent who may be exposed to the same pathogen.

Furthermore, individuals who are immunocompromised are increasingly prone to opportunistic infections.

Immunodeficiency is classified as either primary (congenital), caused by inherited disorders of the immune system, or secondary (acquired), as a result of a disease process or as a result of treatment of other chronic diseases.

Primary causes of immunodeficiency account for more than 200 genetic disorders that impact immune system function. Some examples of primary immunodeficiency include congenital immunodeficiency diseases (e.g., X-linked agammaglobulinemia), combined immunodeficiency disease (e.g., DiGeorge syndrome, Wiskott syndrome), and chronic granulomatous disease.

Examples of secondary immune deficiency include poorly controlled diabetes mellitus defined as HbA1c 9%(0.09) (drawn on admission or within the past three months), end-stage liver disease, hematopoietic malignancies, patients who have had allogeneic hematopoietic stem cell transplantation (HSCT), and patients receiving immunosuppressive therapy (e.g., chemotherapy, radiation, anti-rejection medications, or prolonged high steroid use). Patients with severe, prolonged neutropenia (e.g., absolute neutrophil count less than 500/cu.mm(500x106/L), HSCT, human immunodeficiency virus (HIV) with CD4 count less than 200/cu.mm(200x106/L), a history of splenectomy, splenic dysfunction due to sickle cell disease, and those who have received intense chemotherapy (e.g., childhood acute myelogenous leukemia) are at greatest risk of infection (Centers for Disease Control and Prevention, National Diabetes Statistics Report. 2022; Bonilla et al., J Allergy Clin Immunol 2015, 136: 1186-205 e78; Lipska et al., Diabetes Care 2013, 36: 3535-42).

17:

Medical implants are types of medical devices which enhance, support, or replace existing impaired or missing body parts or functions. These devices may be permanent or temporary and composed of man-made or biologic materials. Indwelling (vs transcutaneous) implants pose a greater risk for the development of cellulitis complicated by bacteremia (U.S. Food and Drug Administration, Implants and Prosthetics. 2023).

18:

Consultation with ophthalmology, neurology, or ear/nose/throat (ENT) specialists should be considered in patients with orbital cellulitis (Hamed-Azzam et al., J Ophthalmic Vis Res 2018, 13: 175-82).

19:

Orbital or post-septal cellulitis affects tissue posterior to the orbital septum, and often results from paranasal sinus disease. Orbital cellulitis is associated with a risk of blindness from damage to the optic nerve (Hamed-Azzam et al.,

J Ophthalmic Vis Res 2018, 13: 175-82).

20:

Orbital cellulitis imaging may include:

- Computed tomography (CT)
- Magnetic resonance imaging (MRI)

21:

Diagnostic imaging can play a pivotal role in determining the extent of orbital cellulitis and evaluating complications (e.g., intraorbital abscess, intracranial involvement). Contrast-enhanced computed tomography (CT) or magnetic resonance imaging (MRI) may be used to diagnose orbital cellulitis. CT is commonly the first imaging study used when an orbital infection is suspected (Kennedy et al., J Am Coll Radiol 2018, 15: S116-s31; Hamed-Azzam et al., J Ophthalmic Vis Res 2018, 13: 175-82).

22:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

23:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

24:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(0.21)

1L 24%(0.24) 4L 36%(0.36)

2L 28%(0.28) 5L 40%(0.40)

3L 32%(0.32) 6L 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)

- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

25:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

26:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

27:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

28:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg

in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

29:

Patients with sepsis-induced tissue hypoperfusion can present with elevated blood lactate levels greater than 2 mmol/L(18.0 mg/dL) (Shankar-Hari et al., JAMA 2016, 315: 775-87).

30:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

31:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

32:

A cellulitis infection that is slow to respond to treatment is characterized by persistent inflammation with leukocytosis, pain, edema, warmth, or an area of increased erythema (Sullivan and de Barra, Clin Med (Lond) 2018, 18: 160-3).

33:

Anti-infective medications may cause adverse drug reactions. Symptoms of an adverse drug reaction include urticaria, angioedema, bronchospasm, pruritus, rash, vomiting, diarrhea, anaphylaxis, and thrombocytopenia.

34:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

35:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

36:**Cellulitis, expected progress after 48h:**

- Afebrile
- Erythema and edema markedly decreased
- Activities at or close to baseline
- Preparing for discharge within the next 24h

If not, consider:

- Infectious disease consult, if not already initiated
- Antibiotic regimen change for persistent fever, leukocytosis, or worsening clinical status

37:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

38:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

39:

Symptoms of inflammation may include:

- Pain
- Edema
- Warmth
- Erythema

40:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

41:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

42:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

43:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g.,

upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

44:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

45:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

46:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

47:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of

the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

48:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

49:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

50:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

51:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included

- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

52:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

53:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

54:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

55:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

56:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

57:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

58:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

59:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

60:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.